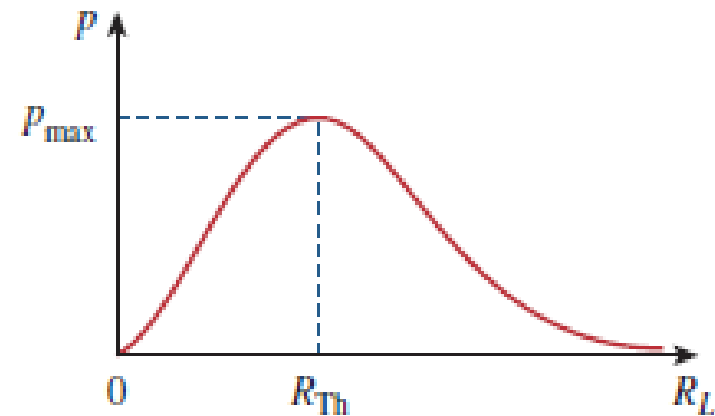
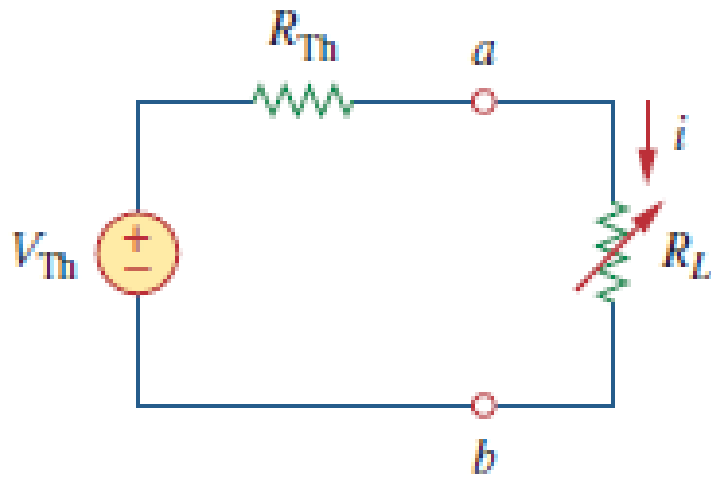


Maximum Power Transfer Theorem

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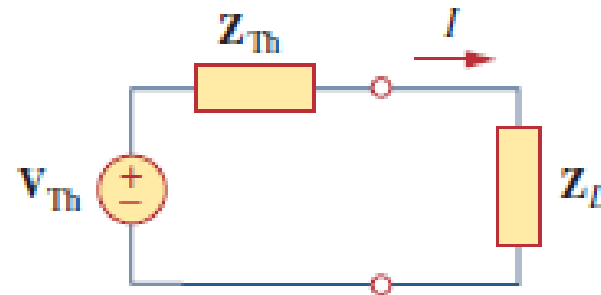
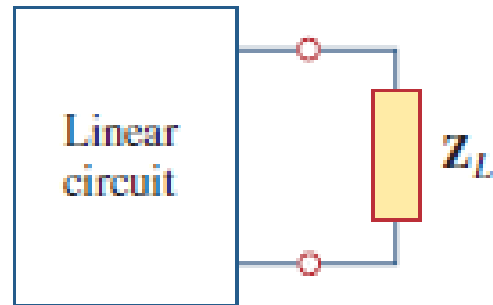
Maximum Power Transfer Theorem

Maximum power is transferred to the load when the load resistance equals the Thevenin resistance as seen from the load ($R_L = R_{Th}$).



$$p = i^2 R_L = \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 R_L$$

Maximum Average Power Transfer



$$Z_{Th} = R_{Th} + jX_{Th}$$

$$Z_L = R_L + jX_L$$

$$Z_L = R_L + jX_L = R_{Th} - jX_{Th} = Z_{Th}^*$$

For maximum average power transfer, the load impedance Z_L must be equal to the complex conjugate of the Thevenin impedance Z_{Th} .

This result is known as the *maximum average power transfer theorem* for the sinusoidal steady state.

Setting $R_L = R_{Th}$ and $X_L = -X_{Th}$

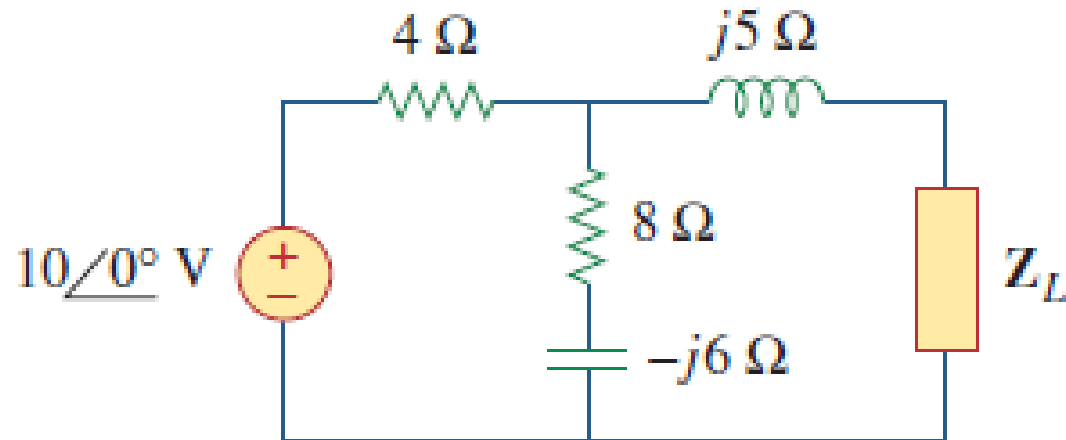
$$P_{\max} = \frac{|V_{Th}|^2}{8R_{Th}}$$

$$R_L = \sqrt{R_{Th}^2 + X_{Th}^2} = |Z_{Th}|$$

Numerical - 1

Determine the load impedance Z_L that maximizes the average power drawn from the circuit

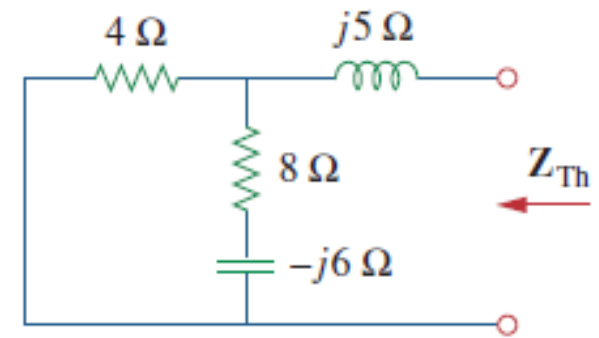
What is the maximum average power?



Solution:

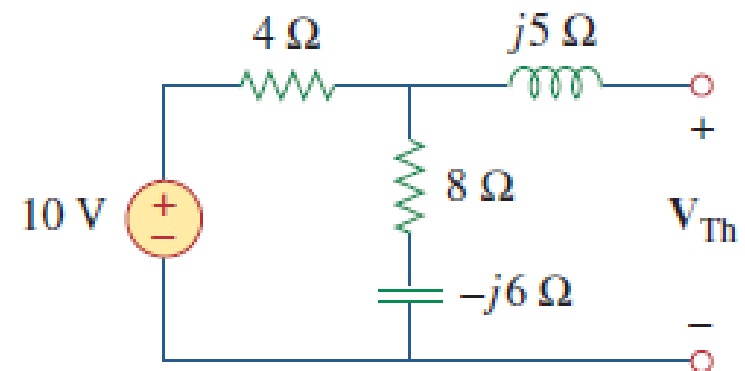
To find Z_{Th} :

$$\begin{aligned} Z_{Th} &= j5 + 4 \parallel (8 - j6) = j5 + \frac{4(8 - j6)}{4 + 8 - j6} \\ &= 2.933 + j4.467 \Omega \end{aligned}$$



To find V_{Th} :

$$V_{Th} = \frac{8 - j6}{4 + 8 - j6}(10) = 7.454 \angle -10.3^\circ \text{ V}$$



The load impedance draws the maximum power from the circuit when

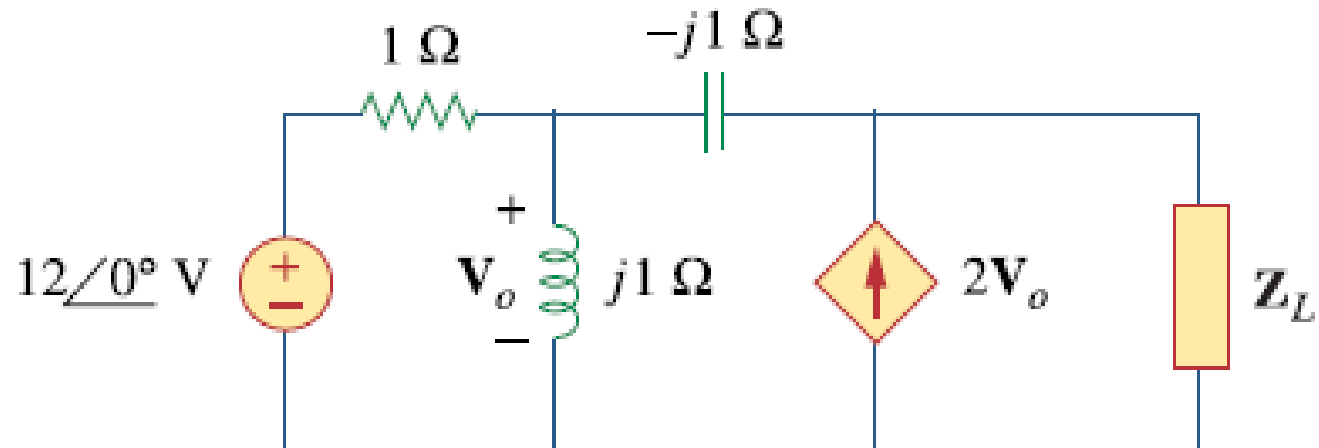
$$Z_L = Z_{Th}^* = 2.933 - j4.467 \Omega$$

the maximum average power is

$$P_{\max} = \frac{|V_{Th}|^2}{8R_{Th}} = \frac{(7.454)^2}{8(2.933)} = 2.368 \text{ W}$$

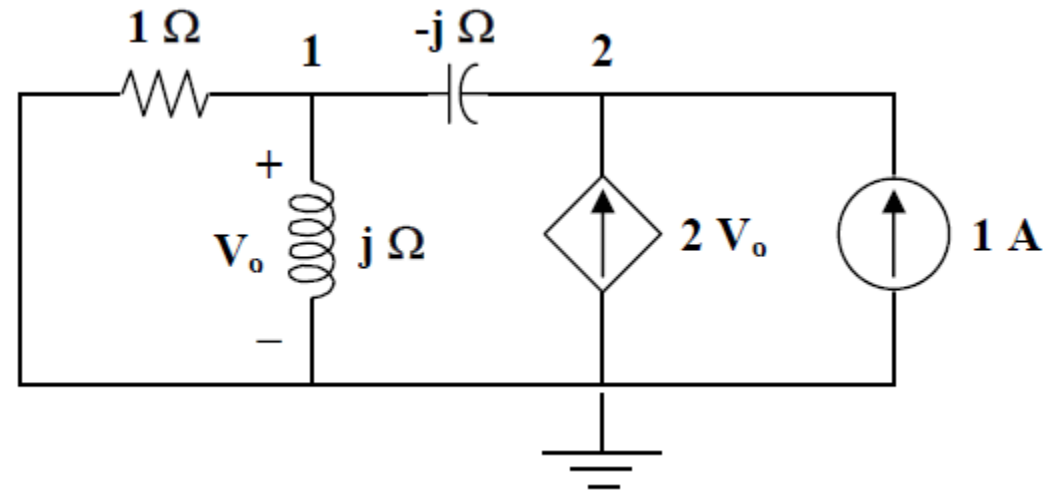
Numerical - 2

In the circuit of Fig. find the value of Z_L that will absorb the maximum power and the value of the maximum power.



- Solution:

To find $\mathbf{Z}_{Th}$, insert a 1-A current source at the load terminals



At node 1,

$$\frac{V_o}{1} + \frac{V_o}{j} = \frac{V_2 - V_o}{-j} \longrightarrow V_o = jV_2 \quad (1)$$

At node 2,

$$1 + 2V_o = \frac{V_2 - V_o}{-j} \longrightarrow 1 = jV_2 - (2 + j)V_o \quad (2)$$

Substituting (1) into (2),

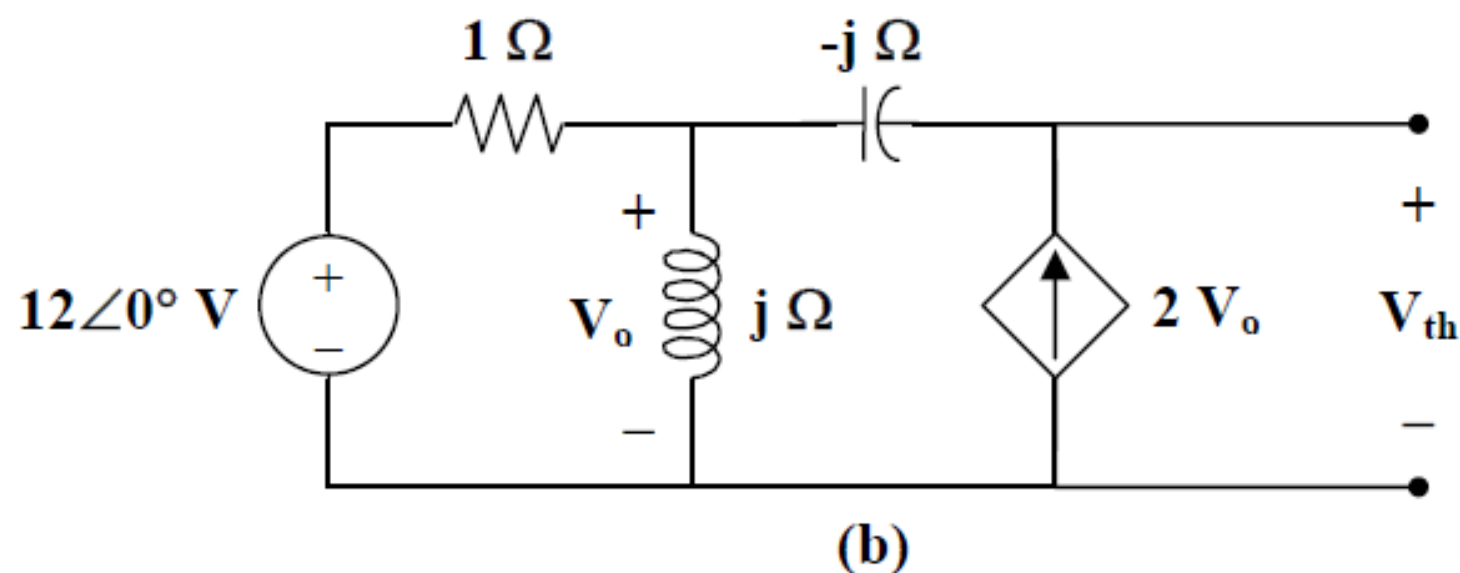
$$1 = jV_2 - (2 + j)(j)V_2 = (1 - j)V_2$$

$$V_2 = \frac{1}{1 - j}$$

$$Z_{Th} = \frac{V_2}{1} = \frac{1 + j}{2} = 0.5 + j0.5$$

$$Z_L = Z_{Th}^* = \underline{\underline{0.5 - j0.5 \Omega}}$$

We now obtain V_{Th} from Fig. (b).



$$2\mathbf{V}_o + \frac{12 - \mathbf{V}_o}{1} = \frac{\mathbf{V}_o}{j}$$

$$\mathbf{V}_o = \frac{-12}{1 + j}$$

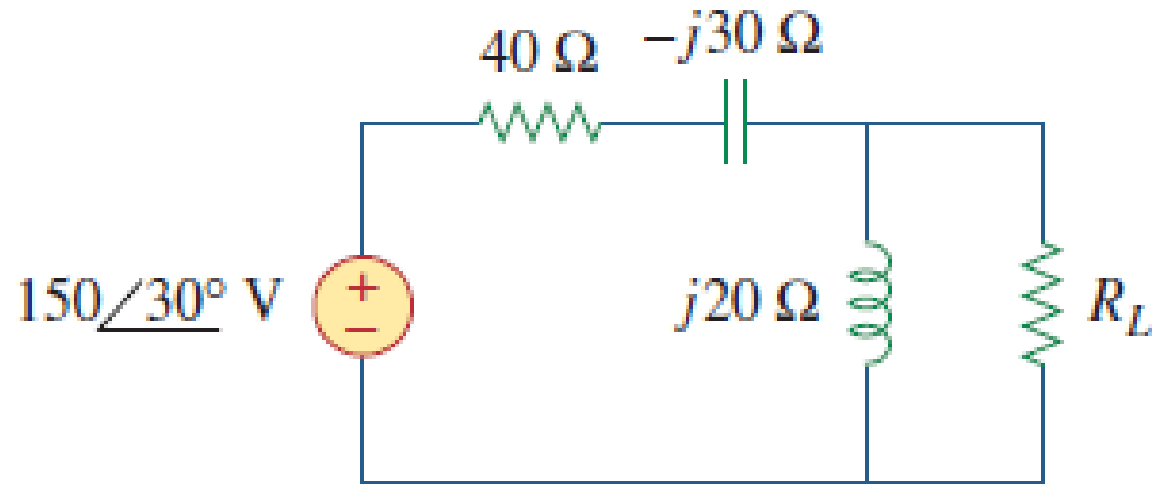
$$-\mathbf{V}_o - (-j \times 2\mathbf{V}_o) + \mathbf{V}_{Th} = 0$$

$$\mathbf{V}_{Th} = (1 - j2)\mathbf{V}_o = \frac{(-12)(1 - j2)}{1 + j}$$

$$P_{\max} = \frac{|\mathbf{V}_{Th}|^2}{8R_L} = \frac{\left(\frac{12\sqrt{5}}{\sqrt{2}}\right)^2}{(8)(0.5)} = \underline{\underline{90 \text{ W}}}$$

Numerical - 3

Find the value of R_L that will absorb the maximum average power.
Calculate the power.



Solution:

$$\mathbf{Z}_{\text{Th}} = (40 - j30) \parallel j20 = \frac{j20(40 - j30)}{j20 + 40 - j30} = 9.412 + j22.35 \, \Omega$$

By voltage division, $\mathbf{V}_{\text{Th}} = \frac{j20}{j20 + 40 - j30} (150 \angle 30^\circ) = 72.76 \angle 134^\circ \text{ V}$

The value of R_L that will absorb the maximum average power is

$$R_L = |\mathbf{Z}_{\text{Th}}| = \sqrt{9.412^2 + 22.35^2} = 24.25 \, \Omega$$

The current through the load is

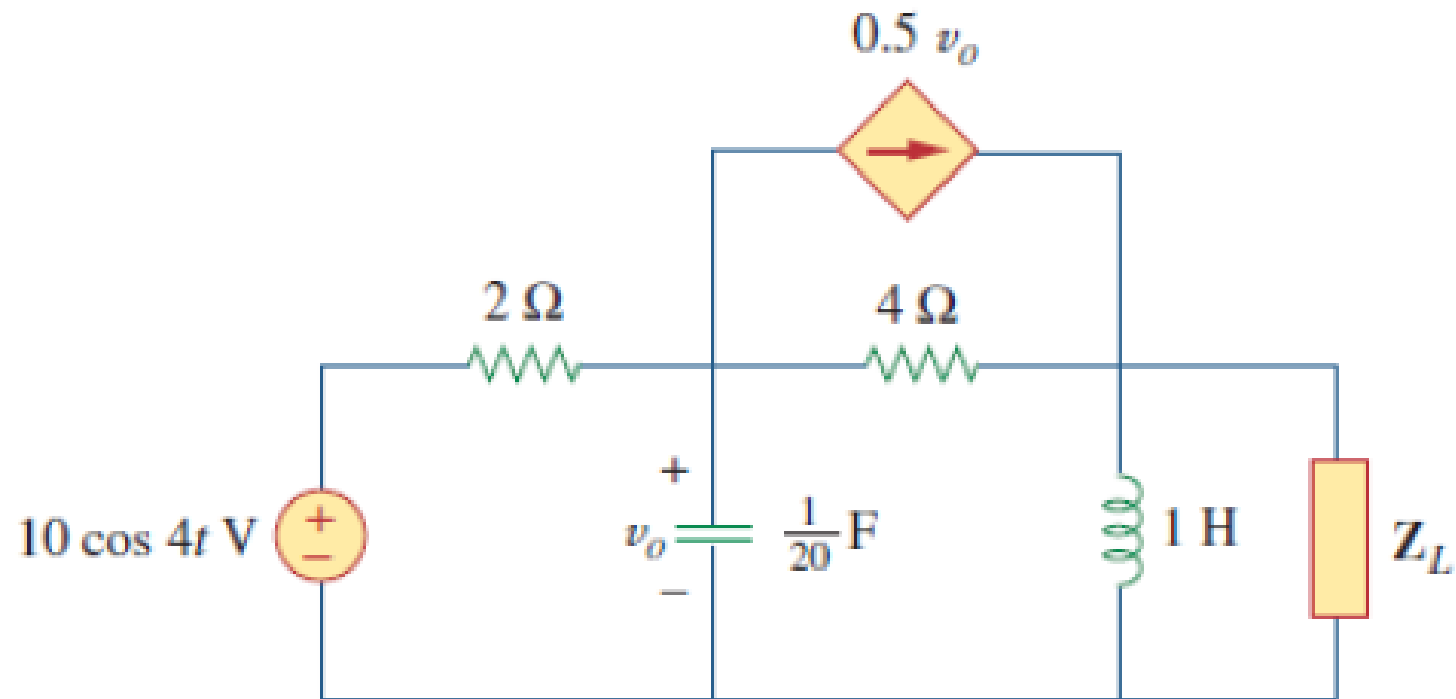
$$\mathbf{I} = \frac{\mathbf{V}_{\text{Th}}}{\mathbf{Z}_{\text{Th}} + R_L} = \frac{72.76 \angle 134^\circ}{33.66 + j22.35} = 1.8 \angle 100.42^\circ \text{ A}$$

The maximum average power absorbed by R_L is

$$P_{\text{max}} = \frac{1}{2} |\mathbf{I}|^2 R_L = \frac{1}{2} (1.8)^2 (24.25) = 39.29 \text{ W}$$

Numerical - 4

Determine the maximum average power delivered to the load for the circuit



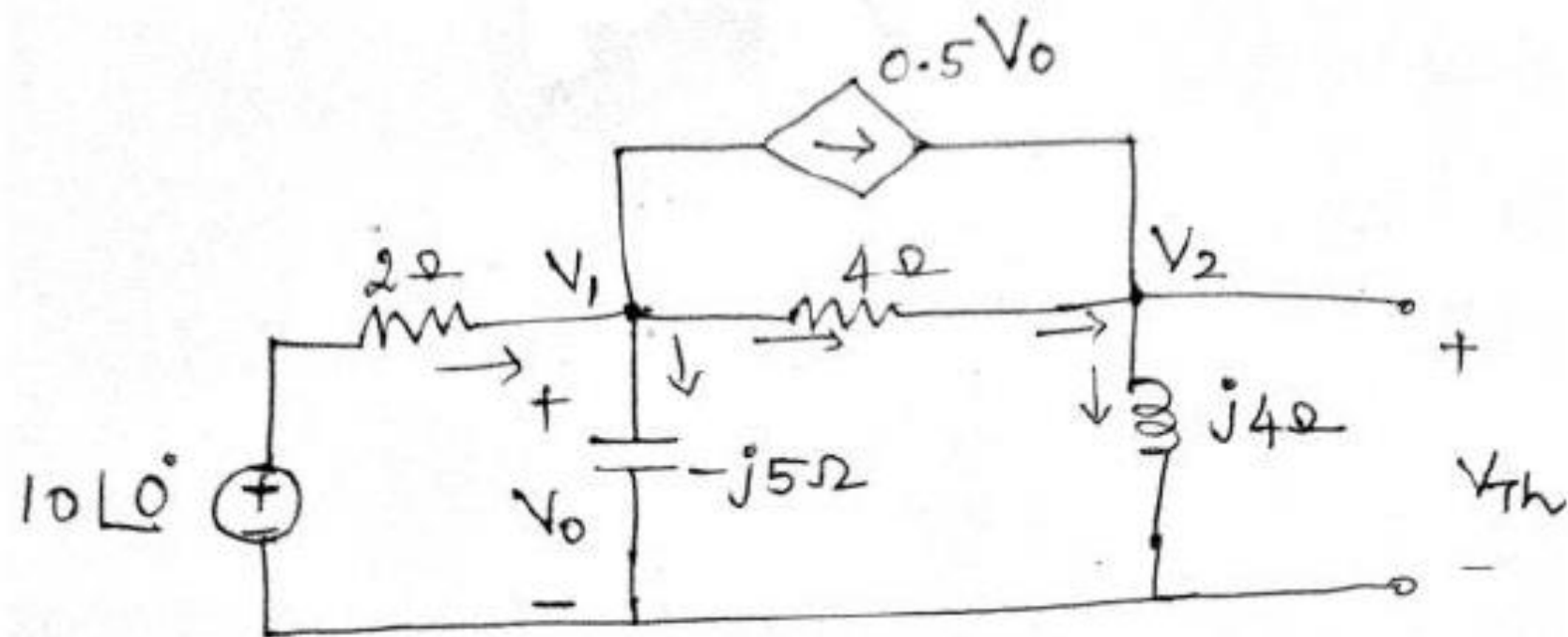
Solution:

$$V_m = 10 ; V = 10 \angle 0^\circ$$

$$\omega = 4 \text{ rad/s}$$

$$1 \text{ H} \Rightarrow j\omega L = j \times 4 \times 1 = j4 \Omega$$

$$\frac{1}{20} \text{ F} \Rightarrow \frac{1}{j\omega C} = \frac{20}{j \times 4} = -j5 \Omega$$



To find V_{th} :-

KCL at node V_1

$$\frac{10 - V_1}{2} = \frac{V_1}{-j5} + 0.5V_0 + \frac{V_1 - V_2}{4} \quad \text{Since } V_0 = V_1$$

$$\frac{10 - V_1}{2} = j0.2V_1 + 0.5V_1 + 0.25(V_1 - V_2)$$

$$5 - 0.5V_1 = j0.2V_1 + 0.5V_1 + 0.25V_1 - 0.25V_2$$

$$1.25V_1 + j0.2V_1 - 0.25V_2 = 5$$

$$\therefore (1.25 + j0.2)V_1 - 0.25V_2 = 5 \rightarrow \textcircled{1}$$

KCL at node V_2 .

$$\frac{(V_1 - V_2)}{4} + 0.5V_1 = \frac{V_2}{j4}$$

$$0.25(V_1 - V_2) + 0.5V_1 = -j0.25V_2$$

$$0.25V_1 - 0.25V_2 + 0.5V_1 + j0.25V_2 = 0$$

$$0.75V_1 + (-0.25 + j0.25)V_2 = 0 \rightarrow \textcircled{2}$$

Solving $\textcircled{1}$ & $\textcircled{2}$

$$\begin{bmatrix} 1.25 + j0.25 & -0.25 \\ 0.75 & -0.25 + j0.25 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$$

$$\Delta = \begin{bmatrix} 1.25 + j0.2 & -0.25 \\ 0.75 & -0.25 + j0.25 \end{bmatrix}$$

$$= (1.25 + j0.2)(-0.25 + j0.25) + 0.25 \times 0.75$$

$$= -0.3125 + j0.3125 - j0.05 - 0.05 + 0.1875$$

$$\Delta = -0.175 + j0.2625 = 0.3154 \angle 123.7^\circ$$

$$\Delta_2 = \begin{bmatrix} 1.25 + j0.2 & 5 \\ 0.75 & 0 \end{bmatrix}$$

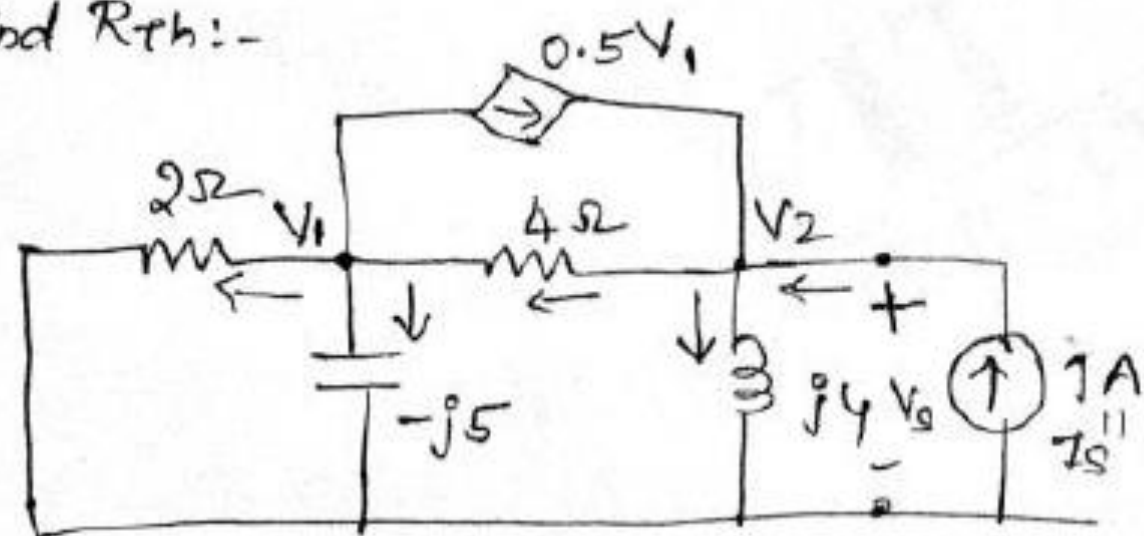
$$\Delta_2 = 0 - 5(0.75) = -3.75 = 3.75 \angle 180$$

$$V_2 = \frac{\Delta_2}{\Delta} = \frac{3.75 \angle 180}{0.3154 \angle 123.7}$$

$$V_2 = 11.88 \angle 56.3^\circ = (6.59 + j9.88) \text{ V}$$

$$V_{Th} = V_2 = (6.59 + j9.88) \text{ V} = 11.88 \angle 56.3^\circ \text{ V}$$

To find R_{th} :-



KCL at node 1,

$$\frac{V_1}{2} + \frac{V_1}{-j5} + 0.5V_1 = \frac{V_2 - V_1}{4}$$

$$0.5V_1 + j0.2V_1 + 0.5V_1 = 0.25(V_2 - V_1)$$

$$1.25V_1 + j0.2V_1 - 0.25V_2 = 0$$

$$(1.25 + j0.2)V_1 - 0.25V_2 = 0 \quad \text{--- (3)}$$

KCL at node 2,

$$1 + 0.5V_1 = \frac{V_2 - V_1}{4} + \frac{V_2}{j4}$$

$$1 + 0.5V_1 = 0.25(V_2 - V_1) - j0.25V_2$$

$$1 = 0.25V_2 - 0.25V_1 - 0.5V_1 - j0.25V_2$$

$$-0.75V_1 + (0.25 - j0.25)V_2 = 1 \longrightarrow \textcircled{2}$$

Solving $\textcircled{3}$ & $\textcircled{4}$

$$\textcircled{4} \Rightarrow 0.75V_1 + (-0.25 + j0.25)V_2 = -1 \longrightarrow \textcircled{4}$$

$$\Delta = 0.3154 \angle 123.7^\circ$$

$$\Delta_2 = \begin{bmatrix} 1.25 + j0.2 & 0 \\ 0.75 & -1 \end{bmatrix}$$

$$= -1.25 - j0.2 = 1.27 \angle -170.9^\circ$$

$$\underline{V}_2 = \frac{\Delta_2}{\Delta} = \frac{1.27 \angle -170.9}{0.3154 \angle 123.7}$$

$$\underline{V}_2 = 4.026 \angle -294.6 = 4.026 \angle 65.4^\circ$$

$$\underline{V}_2 = \underline{V}_g$$

$$\underline{Z}_{Th} = \frac{\underline{V}_g}{\underline{I}_g} = \frac{4.026 \angle 65.4^\circ}{1} = 4.026 \angle 65.4^\circ$$

$$\therefore \underline{Z}_{Th} = 4.026 \angle 65.4^\circ = 1.675 + j3.66 \, \Omega$$

$$\therefore P_{max} = \frac{|\underline{V}_{Th}|^2}{8R_{Th}} = \frac{(11.88)^2}{8(1.675)} = \frac{141.13}{13.4}$$

$$\boxed{P_{max} = 10.53 \text{ W}}$$